



National Institute of Business Management

Internet of Things (Smart Wall Socket)

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Course	:	HND-SE
	:	MAHNDSE261F-001
	:	D.P.D Madushan
Members	:	MAHNDSE261F-007
	:	A.G.J.K Jayathissa

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Project Description

In a day to day life, we consume electricity in many ways, but at the same time we don't know how much electricity we have consumed until the end of the month. The only way of monitoring is either the **Cutout Box** or the **Electricity Bill** itself. This raises a problem, because in a normal house traditional wall sockets do not provide how much electricity is consumed, users cannot track real time electricity usage without buying expensive proprietary smart plugs.

As a solution for this we have come up with **Smart Wall Plugs**, even though they are same as market available smart plugs, our one provides set of new features and most importantly it is affordable. We utilize an ESP-32 with a *non-invasive current sensor* and *voltage module* to calculate the real time wattage and transmitted it into a user-friendly mobile application.

Expected Functionalities

1. Real-Time Electricity Usage

Measures AC Voltage and AC Current to calculate the active power usage (wattage).

2. Remote control the wall socket (switch on / off)

Ability to switch on / off the wall socket remotely via mobile app.

3. Mesh Networking

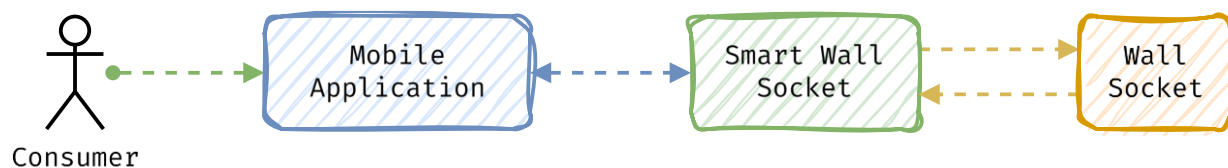
Multiple wall sockets can be configured to a single network, once configured there will be a one master socket which can be used to switch on / off all the other plugs simultaneously.

4. Usage Analytics & Bill Price Calculation

Based on the usage statistics, the used units will be processed through CEB bill calculation algorithm to give estimated electricity bill.

Block Diagram

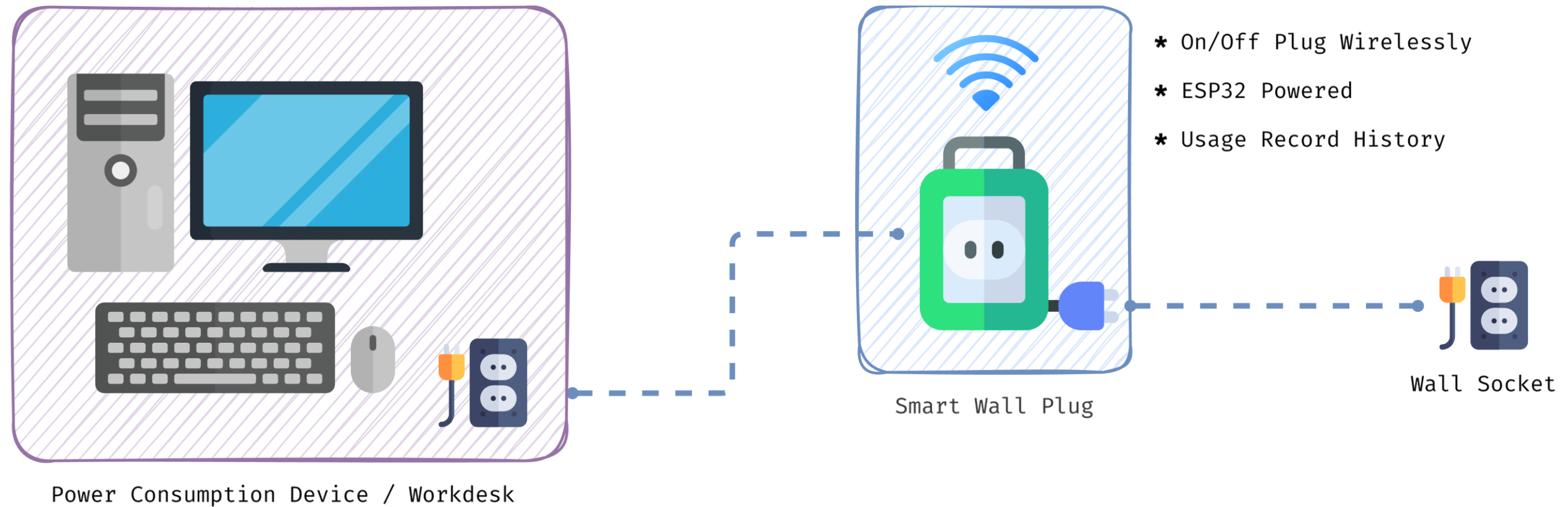
Below is a diagram of how the system is intended to work.

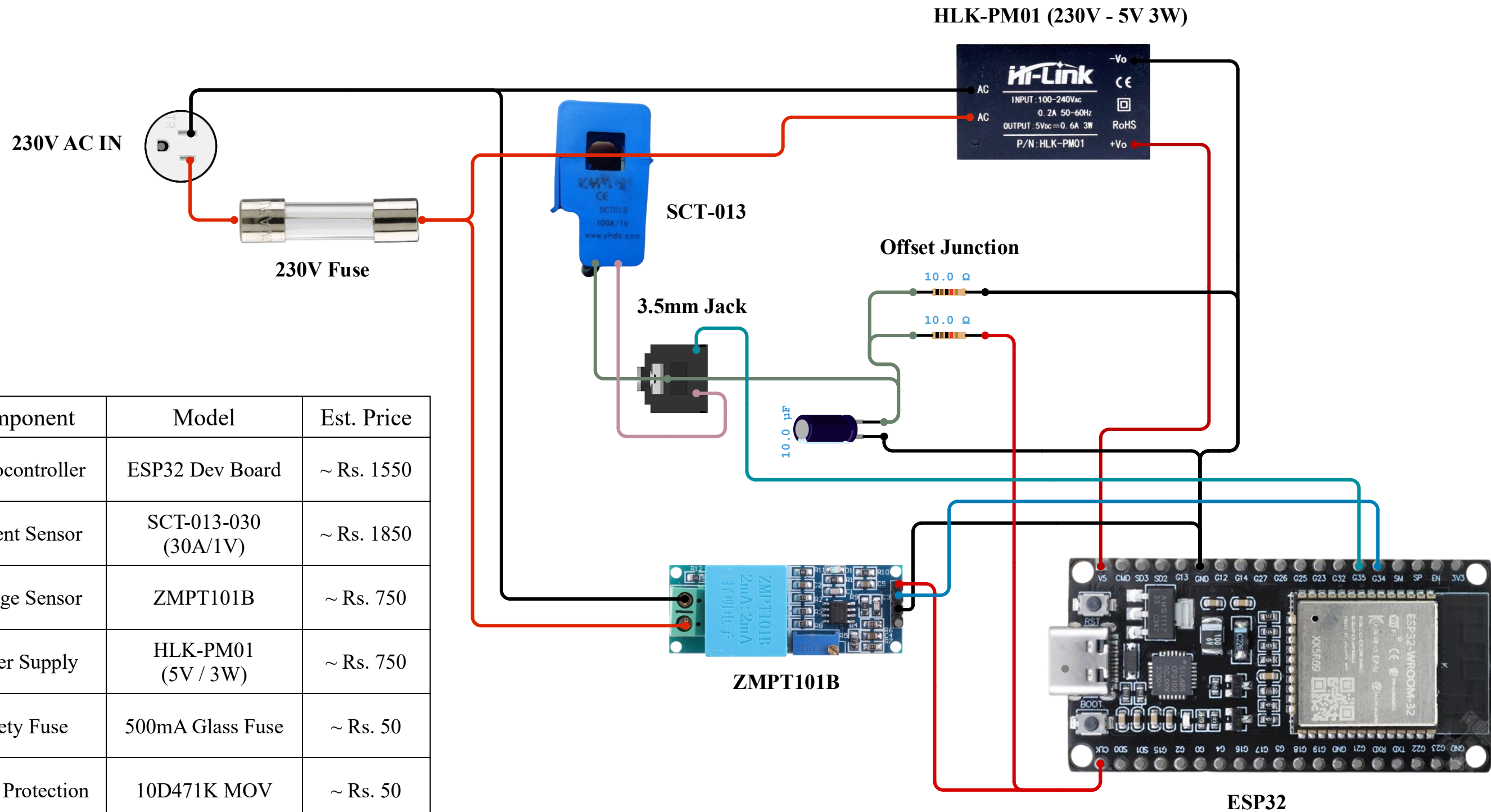


Budget Estimation and Components List

Component / Module	Qty	Source	Price
ESP32 Development Board	1	NIBM	-
Solid State Relay (SSR) Module	1	Tronic.lk	Rs. 340.00
SCT-013-030 non-invasive 30A AC Current Sensor	1	Tronic.lk	Rs. 1850.00
ZMPT101B AC Voltage Sensor Module (2mA)	1	Tronic.lk	Rs. 500.00
Hi-Link HLK-5M05 220VAC to 5VDC 5W	1	Tronic.lk	Rs. 950.00
Glass Fuse & Holder 230VAC 500mA (5x20mm)	2	Tronic.lk	Rs. 60.00
10D471K 470V Metal Oxide Varistor (MOV)	4	Tronic.lk	Rs. 60.00
10k Ω 1/4W Resistors (x2) & 10uF 16V Capacitor (x1) (For DC Bias)	4	Tronic.lk	Rs. 50.00
3.5mm Stereo Female Audio Socket (PCB Mount)	2	Tronic.lk	Rs.60.00
5.08mm Pitch 2-Pin Screw Terminal Block (PCB Mount)	2	Tronic.lk	Rs.60.00
		Approx ~	Rs.4000.00 /~

* Prices are estimated and can be changed if the modules cannot be sources as expected.





Component	Model	Est. Price
Microcontroller	ESP32 Dev Board	~ Rs. 1550
Current Sensor	SCT-013-030 (30A/1V)	~ Rs. 1850
Voltage Sensor	ZMPT101B	~ Rs. 750
Power Supply	HLK-PM01 (5V / 3W)	~ Rs. 750
Safety Fuse	500mA Glass Fuse	~ Rs. 50
Surge Protection	10D471K MOV	~ Rs. 50
Offset Junction	2× 10kΩ Resistors 1× 10μF Capacitor	~ Rs. 50
Connector	3.5mm Female Audio Jack	~ Rs. 150